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What is a Game?

CS 1382 Introduction to Computer Science…

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Warmup

- What kinds of games do we play?
- What is your favorite game (currently)?
- Why do you like that game?
- What sorts of feelings does that game invoke?
- How does the design of that game contribute to those feelings?

What fundamental concepts make a game?

- Goals
- Rules
- Mechanics
- Conflict/Challenge
- Feedback
- Agency
- Uncertainty
- Resolution
- Context
- Interactivity
- Narrative
- Progression
- Social Interaction

How to think like a Game Designer

- Mechanics
  - The actual workings of the games
  - *How much currency do I start with?*
- Dynamics
  - How the player acts in response to mechanics
  - *How do I respond to the currency limitations?*
- Aesthetics
  - How the player feels when acting
  - *How do my actions make me feel?*

MDA Example: Horror Video Game

- Aesthetics
  - How do we want the player to feel?
  - *Tension, scared, careful, suspenseful*
- Dynamics
  - How do we want the player to respond?
  - *Play with caution, carefully, checks everything, hunts for save points*
- Mechanics
  - How should our "save game" feature work? (Select which one seems best)
  - *Auto Save every 2 minutes*
  - *Instant & Invisible Save Game checkpoints*
  - *Save Game stations with a saving wait time*

What steps are involved with making a game?

- 1. Concept Development

  Goal: Identify the core idea of the game

- Key activities:
  - Brainstorm and define the core experience (what you want players to feel or do)
  - Identify the type of game (strategy, puzzle, social, educational, etc.)
  - Determine the audience, player count, and intended platform or medium
  - Write a high-level concept statement or elevator pitch
  - “A cooperative card game where players race to contain a spreading virus before time runs out.”
- Deliverables:
  - Concept statement
  - Preliminary theme and setting
  - Target audience profile
  - Design goals and constraints

What steps are involved with making a game?

- 2. Research and Analysis

  Goal: Ground the concept in context and feasibility

- Key activities:
  - Study similar games (mechanics, player experience, market gaps)
  - Analyze player psychology and motivational models (e.g., Bartle types, Self-Determination Theory)
  - Consider technical, material, or physical constraints (for non-digital games, prototyping supplies; for digital, platform specs)
  - Identify core learning or engagement objectives if educational
- Deliverables:
  - Competitive analysis
  - Reference game library
  - Requirements summary

What steps are involved with making a game?

- 3. Core Mechanics and Rules Design

  Goal: Define how the game works

- Key activities:
  - Design the core loop (the repeated sequence of player actions and feedback)
  - Establish rules, systems, and mechanics that align with your goals
  - Determine win/loss conditions, scoring systems, and progression
  - Balance player agency and randomness
  - Consider fairness, pacing, and player feedback
- Deliverables:
  - Game design document (GDD) or rulebook draft
  - Core mechanics outline
  - Example play scenarios

What steps are involved with making a game?

- 4. Prototyping

  Goal: Quickly test and refine the game's mechanics

- Key activities:
  - Build a low-fidelity prototype (paper cards, placeholder tokens, basic digital mockup)
  - Test for playability, clarity of rules, and fun factor
  - Use rapid iteration: test, adjust, test again
  - Collect qualitative feedback from early players
- Deliverables:
  - Playable prototype
  - Playtest feedback notes
  - Iteration log

What steps are involved with making a game?

- 5. Playtesting and Iteration

  Goal: Validate and refine the player experience

- Key activities:
  - Conduct structured playtesting sessions with diverse player groups
  - Measure engagement, difficulty, balance, clarity, and emotional flow
  - Apply iterative design principles (design → test → analyze → refine)
  - Use quantitative and qualitative data to make changes
  - Ensure rules are intuitive, fair, and strategically rich
- Deliverables:
  - Finalized rulebook
  - Polished prototype
  - Playtesting report and revision history

What steps are involved with making a game?

- ## 6. Aesthetics and Theming

  Goal: Integrate visual, auditory, and narrative elements

- Key activities:
  - Develop art direction, tone, and style guide
  - Refine narrative structure, characters, or world-building
  - Ensure that aesthetics reinforce mechanics and player emotion (MDA alignment)
  - For physical games: finalize component design and layout
  - For digital: develop UI/UX wireframes and interaction flows
- Deliverables:
  - Art and theme guide
  - Narrative script or world summary
  - Component mockups / UI prototypes

What steps are involved with making a game?

- 7. Production

  Goal: Build the final, publishable version

- Key activities:
  - Implement finalized systems, artwork, and polish
  - Conduct balance testing and quality assurance (QA)
  - For digital games: develop code, integrate assets, and test across devices
  - For physical games: create production-ready files and test sample prints
  - Manage schedule, budget, and team workflow (using agile or waterfall models depending on scale)
- Deliverables:
  - Final game or release candidate
  - QA documentation
  - Manufacturing or deployment plan

What steps are involved with making a game?

- 8. Launch and Distribution

  Goal: Deliver the game to players.

- Key activities:
  - Prepare marketing, rule explanations, and tutorials
  - Conduct soft launch or beta if applicable
  - Distribute through appropriate channels (app store, print manufacturer, website)
  - Gather post-launch feedback for patches or revisions
- Deliverables:
  - Published game
  - Instructional materials
  - Launch report and analytics

What steps are involved with making a game?

- 9. Post-Launch Support and Evaluation

  Goal: Sustain and improve the game after release

- Key activities:
  - Collect player analytics, reviews, and community feedback
  - Patch, rebalance, or release expansions as needed
  - Document lessons learned for future projects
  - Assess educational, entertainment, or market success metrics
- Deliverables:
  - Postmortem report
  - Updated design documentation
  - Plans for maintenance or sequels

How do I find a great game idea?

- The Harsh Truth: Ideas Are Cheap
  - Everyone has a “million-dollar game idea”
  - But ideas alone don’t create games - execution does
  - The “idea guy” who won’t build, test, or iterate is often the most frustrating person on a dev team
  - A great idea is worth nothing without the work, design skill, and playtesting to bring it to life
  - Successful game developers are learners and problem-solvers

How do I find a great game idea?

- Myth Busters

  "Your game idea must be completely original, and not based on something else"

  - This idea is nonsense
  - Everything is based on what came before it
  - Your game needs to be interesting, but not uninspired

Ecclesiastes 1:9

What has been is what will be,

and what has been done is what will be done,

and there is nothing new under the sun.

How do I find a great game idea?

- So What Makes a Good Game Idea?
  - A clear core experience - what do you want players to feel?
  - A unique mechanical twist - how is your gameplay different?
  - A reason to exist - why would someone play this instead of what already exists?
  - A vision that can be built - it's better to finish a small, focused game than dream up a giant one you'll never make
- Design iteration
  - Improve as you learn
- Brainstorming
  - Humility
  - Finding good critique and input

Homework

- Read Paper: MDA (Week 09 → Reading)
- Play a board game with at least 1 other person
  - Think about the MDA of the game
    - What is the most significant mechanic of the game?
    - What are the resulting dynamics of that mechanic?
    - How do those dynamics make you feel?
    - How does the interactions with other players affect the gameplay?
- Quiz 04 will be due on Sunday night (all online)